

CODEALPHA CYBERSECURITY INTERNSHIP — TASK 2

Phishing Awareness Training Module

Interactive Web-Based Security Awareness Training with Quiz Assessment

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Internship	Code Alpha — Cybersecurity Internship
Task	Task 2 of 4 — Phishing Awareness Training
Tools Used	HTML5, CSS3, Vanilla JavaScript (no dependencies)
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Repository	Phishing Awareness

Submission Summary

This report documents the design and content of an 8-module interactive phishing-awareness training application built as a single self-contained HTML file. It covers phishing taxonomy, email/website red-flag identification, social engineering psychology, real-world case studies, a defensive playbook, and a scored 6-question knowledge-check quiz.

1. Objective

The goal of Task 2 was to create a presentation or online module focused on phishing attacks — explaining how to recognize phishing emails and fake websites, educating about social engineering tactics, providing best practices, and including real-world examples and interactive quizzes for engagement.

Specific Goals

- Create a presentation or online module focused on phishing attacks.
- Explain how to recognize phishing emails and fake websites.
- Educate about social engineering tactics used by attackers.
- Provide best practices and tips to avoid falling victim.
- Include real-world examples and interactive quizzes for better engagement.

2. Tools & Technologies Used

Component	Technology	Purpose
Structure	HTML5	Semantic content structure across 8 slides
Styling	CSS3 (custom)	Dark cybersecurity-themed design, responsive layout
Interactivity	Vanilla JavaScript	Slide navigation, hover tooltips, live quiz scoring
Fonts	Google Fonts	Share Tech Mono and Exo 2 for UI typography
Delivery	Single .html file	Zero dependencies — runs in any browser offline

3. Module Structure & Content

3.1 Eight-Module Slide Structure

The training is organized as a slide deck with persistent navigation (Prev/Next buttons, clickable progress dots, and a top progress bar). Each module builds on the previous one, moving from statistics and taxonomy through to a hands-on knowledge check.

#	Module	Content
1	Introduction	Key statistics (3.4B phishing emails/day, \$4.9M avg breach cost) and training objectives
2	Types of Phishing	Email phishing, spear phishing, whaling, smishing/vishing, clone phishing, pharming
3	Email Red Flags	Annotated fake PayPal phishing email with hover-to-reveal tooltips on each red flag
4	Fake Websites	URL inspection, the HTTPS-padlock misconception, and free verification tools
5	Social Engineering	Six psychological triggers: Urgency, Fear, Authority, Reciprocity, Social Proof, Liking
6	Real-World Cases	Twitter 2020 Bitcoin scam, DNC 2016 hack, Ubiquiti BEC fraud (\$46.7M), Colonial Pipeline
7	Defence Playbook	Eight actionable habits including MFA, password managers, and reporting procedures
8	Knowledge Check Quiz	6 scored multiple-choice questions with instant explanatory feedback per answer

3.2 Interactive Email Red-Flag Annotation

Module 3 reproduces a realistic phishing email (a fake PayPal account-suspension notice) with six elements wrapped in hover-annotated spans. Hovering over any highlighted phrase reveals a tooltip explaining why it is suspicious — the

typosquatted sender domain "paypal.com" (digit 1 instead of letter l), the artificial 24-hour deadline, and the malicious verification link "paypal-secure-login.ru".

3.3 Quiz Engine

The quiz uses inline JavaScript. Each answer button calls `answer(button, isCorrect, questionId)` which disables all options, highlights correct/incorrect answers, displays a tailored explanation, and reveals a "Next Question" button. After question 6, `showScore()` displays a circular indicator out of 6 with a tiered performance message.

```
// Quiz answer handler (simplified)
function answer(btn, isCorrect, qId) {
  const opts = document.querySelectorAll(`#${qId} .option`);
  opts.forEach(o => o.disabled = true); // lock all options
  if (isCorrect) {
    btn.classList.add("correct");
    quizScore++;
  } else {
    btn.classList.add("incorrect");
    // also highlight the correct option in green
  }
  document.getElementById(`fb-${qId}`).classList.add("show");
  document.getElementById(`next-${qId}`).classList.add("show");
}
```

4. Module Walkthrough & Screenshots

The screenshots below were captured by rendering task2_phishing_awareness.html directly in a Chromium browser at 1280x800 resolution, demonstrating the module exactly as an end-user would experience it.

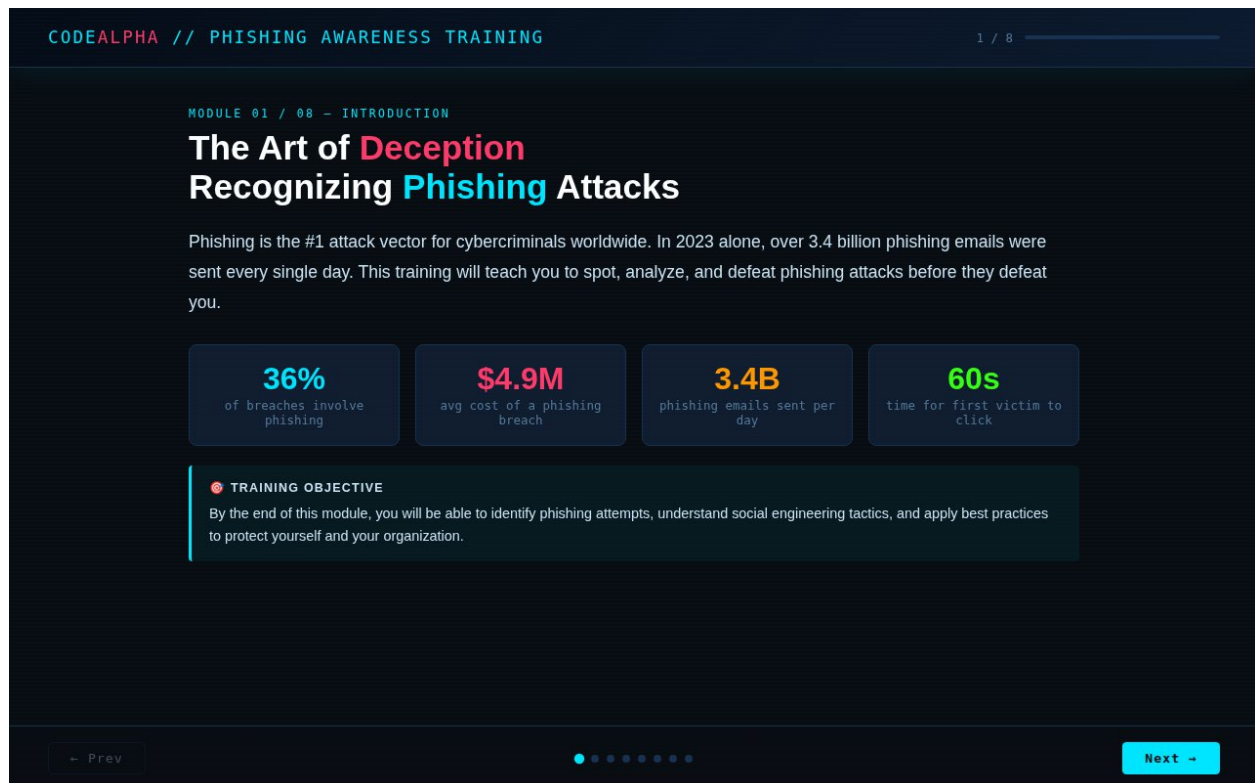


Figure: Module 1 — Introduction slide. Shows the training title, four stat cards (36% of breaches involve phishing, \$4.9M average breach cost, 3.4B phishing emails per day, 60-second time-to-first-click), and the training objective box. The top progress bar and slide indicator (1/8) are visible in the header.

MODULE 03 / 08 - EMAIL RED FLAGS

Anatomy of a Phishing Email

Hover over the highlighted sections below to reveal exactly why each element is suspicious.

▲ PHISHING EXAMPLE - DO NOT CLICK REAL LINKS

From: support@paypa1.com
To: you@yourcompany.com
Subject: ⚠ URGENT: Your account will be SUSPENDED in 24 hours
Date: Mon, 5 Jun 2026 03:14:27 +0000

Dear Valued Customer,

We have detected unusual activity on your account. You must verify your identity immediately or your account will be permanently suspended.

Please click the link below to confirm your details within 24 hours:

👉 <http://paypa1-secure-login.ru/verify/account>

Failure to verify will result in permanent account closure.

Regards,
PayPal Security Team

[← Prev](#)[Next →](#)

Figure: Module 3 — Anatomy of a Phishing Email. A realistic fake PayPal email is rendered with six interactive red-flag annotations (dashed underlines): the typosquatted sender domain "paypa1.com", the urgent subject line, the generic greeting, false urgency in the body, the artificial 24-hour deadline, and the malicious verification link.

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MODULE 05 / 08 — SOCIAL ENGINEERING

The Psychology of Deception

Attackers exploit fundamental human psychology, not just technology. Understanding the six triggers helps you resist them.



Urgency

"Act now or lose access!" — Panic bypasses rational thinking. Slow down when an email demands immediate action.



Fear

"Your account has been compromised!" — Fear triggers fight-or-flight, not critical analysis. Question fear-based messages.



Authority

Impersonating the CEO, IT department, or government agencies. Always verify unusual requests through a separate channel.



Reciprocity

"You've won a prize!" — Creates a sense of obligation. Unsolicited offers are almost always manipulation.



Social Proof

"Millions of users have already updated their info." — Herd mentality pressure. Legitimate services don't pressure this way.



Liking

Pretending to be a colleague or friend to lower your guard. Attackers research social media to craft personalized lures.

GOLDEN RULE

← Prev

Next →

Figure: Module 5 — The Psychology of Deception. Six cards present the core social engineering triggers — Urgency, Fear, Authority, Reciprocity, Social Proof, and Liking — each with an icon, explanation, and real-world phrasing example. A Golden Rule tip box reinforces the core takeaway.

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MODULE 08 / 08 — KNOWLEDGE CHECK

Quick Quiz

Apply what you've learned. 6 questions — let's see how sharp you are.

Question 1 of 6

An email from "support@paypa1.com" warns your account will be suspended. The link leads to "paypa1-secure.ru". What is the PRIMARY red flag?

The email uses urgent language

The sender domain is "paypa1.com" — a typosquat of paypal.com

The email was sent at an unusual time

The email has no image or logo

← Prev

● ● ● ● ● ● ● ●

✓ Complete

Figure: Module 8 — Knowledge Check Quiz, Question 1 of 6. The quiz presents a scenario about a typosquatted PayPal domain and asks the user to identify the primary red flag, with four multiple-choice options rendered as clickable buttons.

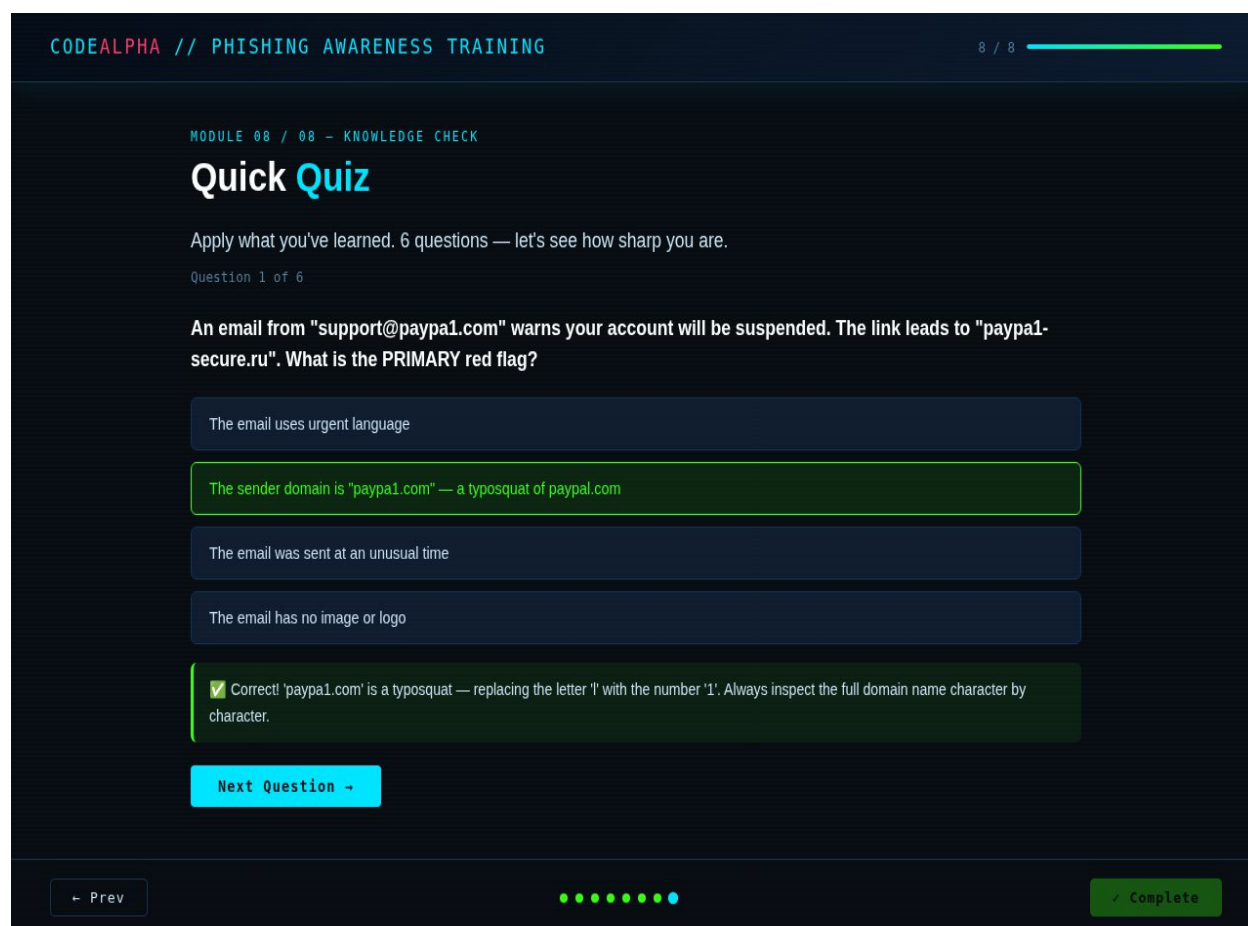


Figure: Quiz feedback in action — after selecting the correct answer (the typosquatted sender domain), the option is highlighted in green, a detailed explanation appears, and a Next Question button becomes available.

5. Results & Analysis

5.1 Quiz Question Bank Summary

Q #	Topic	Correct Answer / Key Concept
1	Email red flags	Typosquatted sender domain (paypa1.com) is the primary indicator
2	HTTPS misconception	HTTPS = encrypted connection only, NOT a guarantee of legitimacy
3	Phishing taxonomy	Personalized, researched attacks = Spear Phishing

Q #	Topic	Correct Answer / Key Concept
4	Best defence	MFA blocks 99.9% of account takeovers even with stolen credentials
5	Incident response	Verify via a known-good channel; never trust contact info from the suspicious message
6	Psychological triggers	Artificial deadlines combine Urgency + Fear manipulation

5.2 Engagement Features

- Progressive disclosure: Content unfolds across 8 modules with a visible progress bar, encouraging completion rather than skipping ahead.
- Hover-to-learn interactions: The red-flag email annotations let users actively explore rather than passively read, improving retention.
- Immediate feedback: Quiz answers are graded instantly with explanations, reinforcing correct mental models even after a wrong guess.
- Tiered scoring feedback: The final score message adapts to performance level, giving users a clear sense of their readiness.
- Zero deployment friction: The single-file HTML design means no installation — runs offline in any browser, or hosted on GitHub Pages in under 5 minutes.

6. Challenges Faced & Learnings

- Balancing realism with safety: The fake phishing email needed to look convincing enough to be educational, while being clearly labeled as a training example to prevent misuse.
 - Single-file constraint: Keeping all CSS, JavaScript, and content in one file required careful organization using CSS variables for theming and clearly separated slide blocks.
 - State management without a framework: Implementing slide navigation, quiz state, and scoring using vanilla JavaScript required explicit DOM manipulation and careful state tracking.
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- Designing effective distractors: Writing plausible-but-incorrect quiz options that test genuine understanding (rather than obviously wrong answers) was important for a meaningful assessment.

7. Conclusion

Task 2 was successfully completed. The resulting phishing awareness training module covers the full CodeAlpha brief: phishing types, email/website red-flag identification, social engineering psychology, four real-world case studies, a defensive playbook, and a scored 6-question quiz with explanatory feedback. The module is self-contained, visually engaging, and requires no installation — making it suitable for direct distribution, GitHub Pages hosting, or corporate security awareness programs.